Below is the **final compiled set** of CS604 for your final preparation — **70 important MCQs** (with answers) and **5 short + 5 long questions** (with answers), fully based on the combined sources so it matches the actual exam trends.

CS604 – Final Term – 70 Important MCQs (with answers)

Operating System Basics

1. OS acts as an interface between: **User and hardware**
2. Multiprogramming increases: **CPU utilization**
3. Time-sharing systems use: **Round Robin scheduling**
4. Kernel is: **Core part of the OS**
5. Batch processing is suitable for: **Non-interactive jobs**

Processes & Threads

6. Process Control Block stores: **Process state, registers, scheduling info**
7. The 5 process states are: **New, Ready, Running, Waiting, Terminated**
8. Context switching saves: **CPU state of a process**
9. Thread is: **Lightweight process**
10. Thread shares: **Code, data, OS resources with other threads**

CPU Scheduling

11. FCFS scheduling is: **Non-preemptive**
12. SJF scheduling gives: **Minimum average waiting time**
13. Round Robin scheduling uses: **Time quantum**
14. Priority scheduling can cause: **Starvation**
15. Aging is used to: **Avoid starvation**

Deadlock

16. Four conditions for deadlock: **Mutual exclusion, Hold & wait, No preemption, Circular wait**
17. Deadlock prevention: **Deny at least one condition**
18. Deadlock avoidance uses: **Banker's algorithm**
19. Resource Allocation Graph detects: **Deadlocks**
20. Wait-for graph is derived from: **Resource allocation graph**

Memory Management

21. Overlay is used to: **Run programs larger than memory**
22. Swapping moves process between: **Main memory and disk**
23. Paging divides memory into: **Fixed-size pages**
24. Page table stores: **Frame number for each page**
25. Segmentation divides memory into: **Variable-sized segments**
26. Bits in segment table entry store: **Protection, valid bit, limit**
27. Page fault occurs when: **Page not in memory**
28. Thrashing occurs due to: **High page fault rate**
29. TLB is: **Translation Lookaside Buffer**
30. Internal fragmentation occurs in: **Paging**

Page Replacement

31. FIFO replaces: **Oldest page**
32. LRU replaces: **Least recently used page**
33. Optimal replaces: **Page not used for longest time**
34. Belady's anomaly can occur in: **FIFO**
35. Clock algorithm uses: **Reference bit**

File Systems

36. Inode stores: **File metadata**
37. Contiguous allocation suffers from: **External fragmentation**
38. Linked allocation suffers from: **Sequential access overhead**
39. Indexed allocation uses: **Index block**
40. Directory stores: **File names and inode numbers**

Disk Management

41. Seek time is: **Time to move disk arm**
42. Rotational latency is: **Time for sector to rotate under head**
43. Disk bandwidth = **(Total bytes transferred / total time)**
44. FCFS disk scheduling: **Processes requests in order of arrival**
45. SSTF chooses: **Nearest track**
46. SCAN moves: **Head back and forth**
47. C-SCAN moves: **One direction, jumps back**
48. Look/CLook are: **Optimized versions of SCAN/C-SCAN**

Synchronization

49. Critical section problem: **Only one process executes at a time**
50. Mutual exclusion ensures: **No two processes in critical section**
51. Semaphore is: **Integer variable for synchronization**
52. Wait(P) decrements semaphore; blocks if: **<0**
53. Signal(V) increments semaphore; wakes waiting process
54. Binary semaphore = **Mutex**
55. Busy waiting wastes: **CPU cycles**
56. Peterson's algorithm solves: **Critical section for 2 processes**
57. Hardware instructions for CS problem: **Test-and-set, Swap, Compare-and-swap, etc.**

IPC & Communication

58. Pipes allow: **Unidirectional communication**
59. Message passing: **send() and receive()**
60. Shared memory: **Multiple processes access same memory**
61. Sockets are used for: **Network communication**

Linux / Practical

62. `fork()` returns: **0 to child, PID to parent**
63. `mknod` creates: **Special files**
64. `cd ~/Desktop` changes directory to: **Desktop**
65. `chmod 755 file` sets: **rwxr-xr-x** permissions
66. `kill -9 PID` sends: **SIGKILL**
67. SIGSTOP: **Stops process, can be resumed**
68. SIGTERM: **Requests process termination**
69. `ps` lists: **Running processes**
70. `top` shows: **Real-time process stats**

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5 Short Questions with Answers

1. Define Page Fault.
Occurs when a process accesses a page not currently in main memory, triggering OS to load it from disk.

2. What are Problems of Overlay?

- Programmer complexity

- Inflexibility
- Increased I/O
- Slower execution
- Manual management of modules

3. What are advantages of using Wait and Signal in semaphores?

- Avoids busy waiting
- Proper process synchronization
- Prevents race conditions

4. Bits in segment table entry for security purpose?

- Valid bit
- Protection bits (Read/Write/Execute)
- Limit (size) field

5. Mention 5 methods to improve hard disk access time and bandwidth.

- Disk scheduling algorithms
- Disk caching
- Shorter seek time drives
- RAID
- Reduce fragmentation

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5 Long Questions with Answers

1. Function to swap two variables.

```
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

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2. Convert Resource Allocation Graph to Wait-for Graph.

- Replace each resource node with edges from holding process to requesting process.
- Example: If P1 holds R1 and P2 requests R1 → P2 waits for P1.

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3. Preempt Resource Deadlock.
Deadlock prevention by forcibly taking resource from a process and giving to another — possible in resources that can be saved/restored (CPU, memory).

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4. 5 Hardware instructions to solve Critical Section Problem.

- Test-and-Set
- Compare-and-Swap
- Exchange (XCHG)
- Fetch-and-Add
- Disable/Enable interrupts

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5. Compare Stack, Hash Table, Sequential Search in terms of performance.

- **Stack:** LIFO, O(1) push/pop, not for search
- **Hash Table:** O(1) average search, unordered
- **Sequential Search:** O(n) search, simple but slow for large data

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